

High-Dimensional Profiling of MVA-Based HIV Vaccine Responses Reveals Dynamic T-Cell Phenotypic Maturation and Establishment of Multifunctional Memory Populations

Keywords

Vaccine-induced immune dynamics|T-cell phenotypic heterogeneity| T-cell memory differentiation| Mass cytometry (CyTOF)| Unsupervised clustering| Longitudinal immunophenotyping

Yaroslava Shevchenko

Margaux Beze

Shubhdeep Kaur

Abstract

Background

Understanding the quality and dynamics of vaccine-induced T-cell responses remains a central challenge in HIV vaccine development. Beyond magnitude alone, the phenotypic diversity, functional capacity, and temporal evolution of antigen-specific T cells are key determinants of protective immunity. Modified Vaccinia Ankara (MVA)–based HIV vaccines are known to elicit robust cellular responses, yet a comprehensive, high-dimensional characterization of these responses across prime and boost immunizations is still lacking.

Methods

Peripheral blood T cells collected longitudinally before and after prime and booster immunizations were profiled using multiparametric flow cytometry. A panel capturing lineage, activation, proliferation, survival, and effector function markers was analyzed using unsupervised computational approaches. Dimensionality reduction (t-SNE and UMAP) and k-means clustering were applied to define phenotypically distinct T-cell populations, followed by quality control, cluster merging, and heatmap-based phenotypic annotation. Statistical comparisons assessed temporal changes in cluster abundance.

Results

Unsupervised analysis revealed extensive phenotypic heterogeneity within vaccine-responsive CD4⁺ and CD8⁺ T-cell compartments, organized into a dense central manifold and several discrete peripheral regions. Peripheral clusters displayed high cell density and distinct phenotypes, including proliferative (Ki67⁺), activated (CD154⁺), and effector-like populations. A large peripheral megacluster characterized by elevated Bcl-2, CD28, IL-10, IL-4, Granzyme B, and CD4 expression was identified and showed dynamic expansion following vaccination, consistent with a vaccine-responsive T-cell population.

Conclusions

This study provides a high-resolution, systems-level view of MVA-induced T-cell immunity, revealing coordinated phenotypic programs associated with vaccine responsiveness. By integrating unsupervised clustering with longitudinal analysis, we identify proliferative, survival-associated, and multifunctional T-cell populations that evolve across prime–boost immunization. These findings advance the understanding of cellular immune signatures elicited by HIV vaccines and provide a scalable framework for evaluating T-cell quality in future vaccine trials.

Introduction

The induction of robust and durable T cell immunity is a central objective of modern vaccine design, particularly for pathogens such as HIV, for which protective antibody responses alone have proven insufficient (5). CD4⁺ and CD8⁺ T cells play complementary roles in antiviral immunity: CD4⁺ T cells provide essential help to both B cells and cytotoxic T lymphocytes, while CD8⁺ T cells directly eliminate infected cells through cytotoxic effector mechanisms such as perforin and granzyme release (1). Vaccine-induced T cell responses are highly

dynamic and typically progress through distinct phases, including **early activation and clonal expansion**, a **contraction phase** characterized by the loss of short-lived effector cells, and the **establishment of long-lived memory populations** capable of rapid recall upon antigen re-exposure (2). Prime–boost vaccination strategies are commonly used to enhance these responses, as a priming immunization establishes antigen-specific T cell populations that can be efficiently re-expanded and functionally enhanced by subsequent booster doses (3). Viral vectors such as **Modified Vaccinia Ankara (MVA)** are particularly effective at inducing strong cellular immune responses and have therefore been widely explored in HIV vaccine research (4, 5). Despite these advances, the phenotypic complexity and temporal evolution of vaccine-induced T cell responses remain incompletely understood, particularly at the single-cell level.

High-dimensional immune profiling has shown that vaccine-elicited T cell responses are heterogeneous and consist of multiple phenotypically and functionally distinct subpopulations, including activated effector cells, proliferating cells, cytotoxic subsets, and regulatory or self-limiting populations (6). Understanding their dynamics is critical for evaluating vaccine efficacy and identifying immune signatures associated with durable protection. In this context, the present study addresses **two main questions**: **(i)** what is the phenotypic diversity of T cell populations induced by an MVA-based HIV vaccine, as defined by activation markers, cytokine production, cytotoxic molecules, and proliferation or survival markers; and **(ii)** how do the abundances and phenotypic characteristics of these populations change across different stages of the immune response, specifically following the prime and booster immunizations.

To address these questions, a longitudinal vaccination study was conducted in five HIV-negative macaques immunized with an MVA vector expressing HIV antigens as a prophylactic strategy to induce *de novo* immune responses in the absence of infection. Blood samples were collected at multiple time points following the prime and booster immunizations to examine the temporal dynamics of vaccine-induced T cell responses. Peripheral blood mononuclear cells were analyzed using mass cytometry (CyTOF) with an extensive T cell–focused marker panel (Table 1). To extract biologically meaningful patterns from these high-dimensional datasets, bioinformatic analyses were performed. Together, this integrated experimental and computational strategy enables the characterization of T cell phenotypic diversity and provides a framework to assess how prime–boost vaccination shapes the magnitude, quality, and longevity of cellular immune responses.

Materials and methods

Vaccination Study Design, Sample Collection and Data acquisition. Five HIV-negative macaques were vaccinated with a Modified Vaccinia Ankara (MVA)–based HIV vaccine using a prime–boost regimen, with the booster administered two months after the priming immunization. Peripheral blood samples were collected longitudinally to assess vaccine-induced immune dynamics. Samples were obtained at baseline prior to vaccination (day –19 relative to prime), and at days 3, 8, 14, and 57 following the prime immunization. Following the booster vaccination, additional blood samples were collected at days 3, 8, 14,

and 28 post-boost (**Fig. 1**). Mass cytometry (CyTOF) using a T cell–focused antibody panel (Table 1) was used for high-dimensional single-cell profiling.

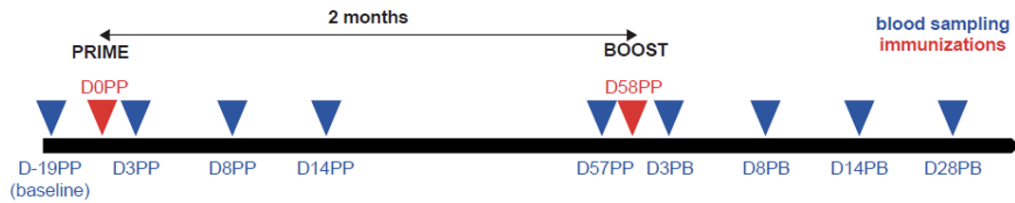


Figure 1. Vaccination Study Design

Table 1. T cell–focused marker panel for mass cytometry analysis.

Description	Markers
Lineage markers	CD3, CD4, CD8
Activation markers	CD69, CD154, HLA-DR
Cytokines	IL-2, IL-4, IL-10, IL-17, TNF α , IFN γ
Cytotoxic molecules	Granzyme B, Perforin
Co-stimulatory receptors	CD27, CD28
Proliferation or survival markers	Ki67, Bcl-2

Data pre-processing and quality control. FCS files were imported into R using the **FlowCore** package. Signal intensities for all markers were transformed using an **arcsinh transformation** to stabilize variance and improve comparability across markers with different dynamic ranges. All samples were merged into a single data frame with sample identifiers retained as metadata. To reduce computational burden while preserving population structure, a uniform random down-sampling strategy was applied to obtain 25,000 cells across samples. Technical and non-informative channels (e.g., time, beads, dead cell markers, and selected lineage markers not relevant to the analysis) were excluded prior to dimensionality reduction and clustering. Quality control procedures included: evaluation of marker expression ranges to identify potential staining or acquisition artifacts, and cell count distributions across samples to detect outliers. Small clusters with fewer than 50 cells and clusters showing excessive uniformity were flagged using dedicated CellVizR quality control functions.

Dimensionality Reduction Using t-SNE and viSNE. To visualize high-dimensional T cell phenotypic structure, t-distributed Stochastic Neighbor Embedding (t-SNE) was applied using the **Rtsne** package. This non-linear dimensionality reduction method preserves local neighborhood relationships between cells and is particularly suited for identifying phenotypically similar immune cell populations. The viSNE approach, conceptually based on t-SNE, was implemented by projecting transformed marker expression values into

two-dimensional space. Resulting embeddings were visualized using **ggplot2**, with cells colored by sample, condition, or marker expression intensity.

CellVizR Framework for Integrated High-Dimensional Analysis. To enable systematic clustering, visualization, and statistical comparisons, data were additionally analyzed using the **CellVizR** framework. FCS files were imported into a DataCell object with arcsinh transformation applied internally. A uniform down-sampling method was used to ensure balanced representation of samples. Manifold Approximation and Projection (UMAP) was used to generate a low-dimensional manifold representation of T cell phenotypes. The resulting manifold served as the basis for unsupervised clustering and visualization of cell density, cluster identity, and marker expression patterns across vaccination stages.

Unsupervised Clustering and Metacluster Definition. Unsupervised clustering was performed on the UMAP manifold using k-means clustering, with 100 and 200 clusters explored to capture both fine-grained and broader phenotypic structures. Cluster abundance distributions were examined across samples and timepoints. Biologically related clusters exhibiting similar marker expression profiles were merged into higher-order metaclusters to represent coherent T cell phenotypes. These metaclusters were used for downstream interpretation and statistical testing.

Phenotypic Characterization of Cell Clusters. Cluster and metacluster phenotypes were characterized by overlaying marker expression intensities onto the UMAP manifold and by generating heatmaps summarizing median marker expression per cluster. This approach enabled identification of activated, cytokine-producing, cytotoxic, or proliferative T cell populations induced by vaccination. Comparisons between post-prime and post-boost samples highlighted phenotypic shifts associated with secondary immune responses. To assess dynamic changes in T cell populations, cluster and metacluster abundances were compared across timepoints and experimental conditions. Statistical testing was performed using unpaired t-tests to compare test conditions (e.g., post-prime or post-boost) against relevant reference conditions (e.g., baseline or earlier timepoints). Results were visualized using volcano plots and heatmaps of p-values to identify significantly expanding or contracting cell populations. Longitudinal trends in cluster and metacluster abundances were visualized using boxplots stratified by timepoint. Timepoints were ordered to reflect the temporal sequence of vaccination and sampling. These analyses enabled quantitative assessment of how specific T cell populations evolved over the course of the immune response, providing insight into the kinetics of vaccine-induced cellular immunity.

Results

t-SNE/viSNE overview of global T-cell structure

The first objective was to obtain an unsupervised, global view of the phenotypic landscape of gated T cells across all macaques and timepoints, and to verify whether samples

segregated strongly (suggesting major condition effects or technical/batch structure) before proceeding to clustering and differential abundance analyses. After arcsinh transformation of marker intensities and removal of technical/non-informative channels (e.g., Time, cell length, beads, dead-cell and other excluded markers), a uniform random subset of 25,000 CD4⁺ T cells was selected from the pooled dataset to generate a two-dimensional t-SNE embedding using the Rtsne implementation (dims = 2, fixed seed). In this representation, each point corresponds to a single cell and spatial proximity reflects similarity in multivariate marker expression patterns (i.e., cells close together have more similar marker profiles than cells farther apart). When the embedding was colored by sample identity (**Fig. 1**), cells from different macaques and timepoints largely overlapped throughout the same continuous manifold, rather than forming distinct sample-specific “islands.” Visually, the t-SNE map displayed a dominant elongated high-density structure with extensive color mixing, indicating that the main T-cell phenotypic states were shared across animals and sampling times in the pooled projection.

To examine whether prime- and boost-associated timepoints altered the global distribution, the same t-SNE coordinates were visualized after grouping samples by experimental condition/timepoint. Across conditions, the overall geometry of the embedding remained comparable, with no obvious condition forming an entirely separate region of the map. Marker-overlay views (coloring by expression intensity on the t-SNE coordinates)(**Supplementary Figure 1**) were used to localize where individual markers were enriched, distinguishing regions of high versus low expression within the embedding and providing an initial, qualitative mapping between areas of the t-SNE space and marker-defined phenotypes (e.g., localized versus diffuse expression patterns across the manifold). Overall, the t-SNE/viSNE visualization provided a shared low-dimensional “atlas” of T cells in which samples largely co-localized, supporting downstream analyses aimed at identifying phenotypically distinct subpopulations within the common structure and quantifying subtle abundance/phenotype shifts across prime and boost timepoints rather than gross sample-to-sample separation.

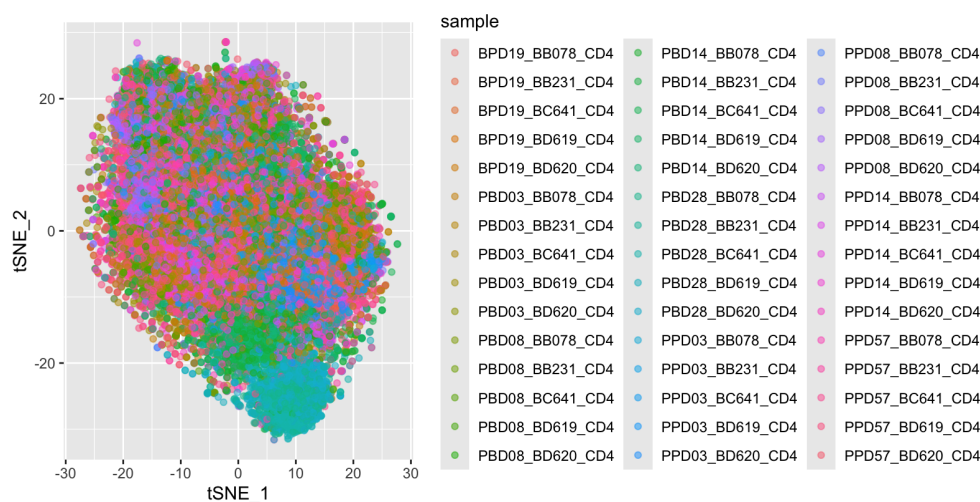


Figure 1. Global t-SNE embedding of pooled T cells across all samples and timepoints. Two-dimensional t-SNE projection of arcsinh-transformed single-cell marker expression for pooled T cells (CD3⁺), generated from a uniform random down-sampling of 25,000 events. Each point represents one cell and is colored by sample identity (macaque × timepoint). Extensive overlap of samples across the embedding indicates a shared phenotypic space across the dataset.

Quality control of mass cytometry data and robustness of unsupervised clustering

We next assessed data quality, marker signal consistency, and the robustness of unsupervised clustering prior to downstream phenotypic characterization and differential abundance analyses. To evaluate signal scaling and inter-sample consistency, individualized marker expression ranges were examined across all samples (**Fig. 2A**). For each marker, median expression levels together with the 5th and 95th percentiles of MFI/MSI values were computed. Lineage and co-stimulatory markers, including CD3, CD4, CD8, CD27, CD28, and CD45RA, exhibited stable and comparable expression ranges across samples, indicating consistent staining and signal acquisition. In contrast, activation and functional markers such as TNF α , IFN γ , Granzyme B, Ki67, and HLA-DR showed broader upper expression ranges, reflecting heterogeneous and inducible expression patterns rather than technical variability. As expected, technical channels (e.g., Cells, Cell length, Dead, Beads) displayed distinct distributions and were excluded from downstream analyses. Following dimensionality reduction, T cells were clustered in manifold space using k-means clustering with $k = 200$ to achieve high-resolution partitioning of phenotypically related cell states. Cluster size quality control was then performed using a minimum threshold of 50 cells per cluster (**Fig. 2B**). Under this criterion, no clusters were classified as small, corresponding to 0% of clusters falling below the size threshold. This result indicates that the selected clustering resolution did not lead to over-segmentation or the generation of poorly supported clusters. Cluster phenotypic coherence was further evaluated using uniform cluster quality control based on intra-cluster variability metrics (**Fig. 2C**). Statistical testing identified clusters with excessive heterogeneity using a combined criterion of $P < 0.05$ and interquartile range (IQR) > 2 across clustering markers. The majority of clusters (84%) met the uniformity criteria, indicating internally coherent marker expression profiles, whereas a minority of clusters showed evidence of increased intra-cluster variability. Together, the assessment of marker expression ranges, cluster size distribution, and intra-cluster uniformity demonstrates high data quality and appropriate clustering granularity, supporting the robustness of the unsupervised clustering strategy and its suitability for downstream phenotypic and longitudinal analyses.

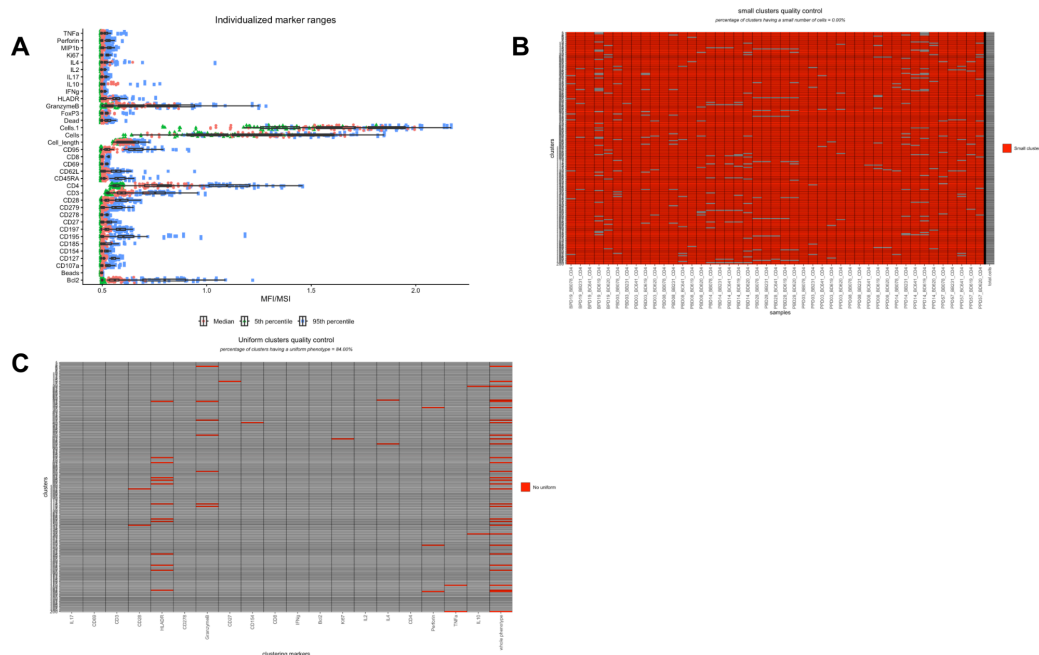


Figure 2. Quality control of mass cytometry data and unsupervised clustering. (A) Individualized marker expression ranges across samples, shown as median (black), 5th percentile (green), and 95th percentile

(blue) MFI/MSI values for each marker. **(B)** Small cluster quality control based on a minimum cluster size threshold of 50 cells, indicating the absence of under-represented clusters. **(C)** Uniform cluster quality control assessing intra-cluster phenotypic variability across clustering markers using statistical criteria ($P < 0.05$; IQR > 2).

UMAP-based manifold structure and identification of phenotypically distinct T-cell clusters

Next, this analysis focused on resolving the phenotypic structure of T cells at high resolution using manifold learning and unsupervised clustering, and to identify phenotypically distinct cell populations occupying discrete regions of the UMAP space. A two-dimensional UMAP embedding was generated from arcsinh-transformed marker expression values to visualize phenotypic similarity among single T cells. The density map revealed a large central manifold region together with several high-density peripheral regions that formed compact “islands” separated from the main body of cells (**Fig. 3A**). Unsupervised clustering was then performed directly in manifold space using k-means clustering with $k = 200$, resulting in a fine-grained partition of the UMAP space into spatially contiguous clusters (**Fig. 3B**). Multiple clusters coincided with the peripheral high-density islands, including clusters 21, 55, and 28 in the upper region of the manifold, a compact block of clusters including 88, 111, 96, etc., and an additional highly separated clusters 176 and 49. A second clustering solution with $k = 100$ preserved the overall manifold structure (Supplementary Fig. 2), supporting that the identification of these discrete regions was not dependent on a single clustering resolution. Overall, UMAP analysis revealed a structured T-cell phenotypic landscape that includes several high-density, spatially separated regions, and k-means clustering identified discrete clusters corresponding to these dense islands, prioritizing them for marker-based phenotypic characterization.

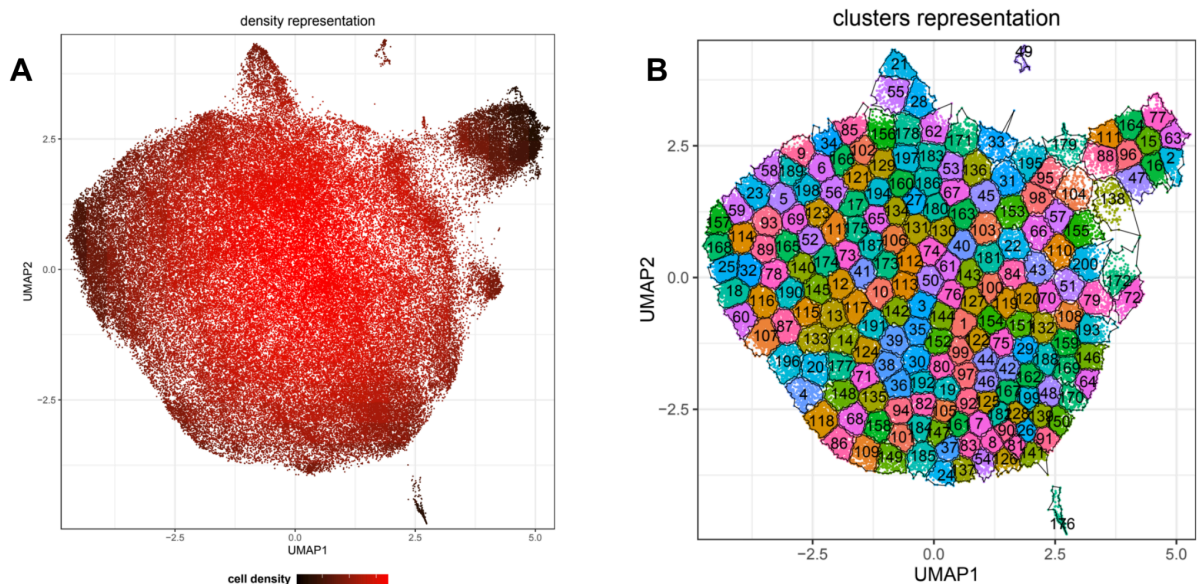


Figure 3. UMAP manifold and k-means clustering of T-cell phenotypic space. **(A)** Density representation of single T cells projected onto the two-dimensional UMAP manifold. Each point represents one cell, and color intensity indicates local cell density. **(B)** Unsupervised k-means clustering ($k = 200$) performed in UMAP manifold space and visualized on the same UMAP coordinates. Each colored region corresponds to an individual cluster, with cluster boundaries separating phenotypically similar groups of cells.

Heatmap-based phenotypic characterization

To determine whether these spatially distinct regions (**Fig. 4A**) corresponded to biologically meaningful T-cell states, we next examined their phenotypic profiles using marker expression heatmaps (**Fig. 4B**). Several spatially segregated clusters displayed distinct and coherent expression patterns, supporting their annotation as biologically relevant T-cell states. Clusters within **Region I** were characterized by elevated expression of the proliferation marker **Ki67**. In particular, clusters 21 and 55 showed strong Ki67 expression with limited expression of cytotoxic molecules, indicating actively cycling T cells (7). Cluster 28, also located in Region I, exhibited high Ki67 expression together with detectable **CD278 (ICOS)** and moderate **Granzyme B** levels, suggesting a proliferative population with emerging effector features (8). Collectively, Region I clusters were distinguished by a proliferative phenotype with variable acquisition of functional markers. A small but clearly separated cluster, cluster 49 (**Region II**), was uniquely characterized by high expression of **CD154 (CD40L)**. This phenotype was not broadly shared with neighboring clusters, indicating a discrete population potentially enriched for antigen-responsive helper T cells (8). At the lower extremity of the UMAP manifold, cluster 176 (**Region IV**) formed a narrow and isolated projection. This cluster was defined by concurrent expression of **CD28** and **Ki67**, indicating a proliferating population retaining co-stimulatory capacity. Its separation from the central manifold and distinct marker profile supported its treatment as a unique phenotypic entity.

In contrast, **Region III** comprised a group of spatially adjacent clusters forming a larger, cohesive peripheral structure. Individual clusters within this region shared overlapping phenotypic features, motivating their aggregation into a single composite population for downstream analyses. Therefore, we merged them into a single metacluster to enable robust quantification of their temporal dynamics across vaccination time points (**Fig. 4C**). Phenotypic profiling of the Region III metacluster showed coordinated upregulation of Bcl-2 and CD28, together with increased expression of CD4, indicating a CD4-enriched population with enhanced survival-associated and co-stimulatory signaling potential (9). In addition, the metacluster displayed increased expression of IL-10 and IL-4, consistent with a cytokine-producing phenotype alongside elevated Granzyme B, indicating concurrent expression of a cytotoxic effector molecule within this CD4-skewed compartment (10). Together, this combination of elevated **CD28/Bcl-2** (co-stimulation/survival), **IL-10/IL-4** (cytokine production), and **Granzyme B** (effector molecule) distinguishes the Region III metacluster from the central manifold and supports its designation as a phenotypically distinct vaccine-responsive T-cell population for longitudinal abundance analyses.

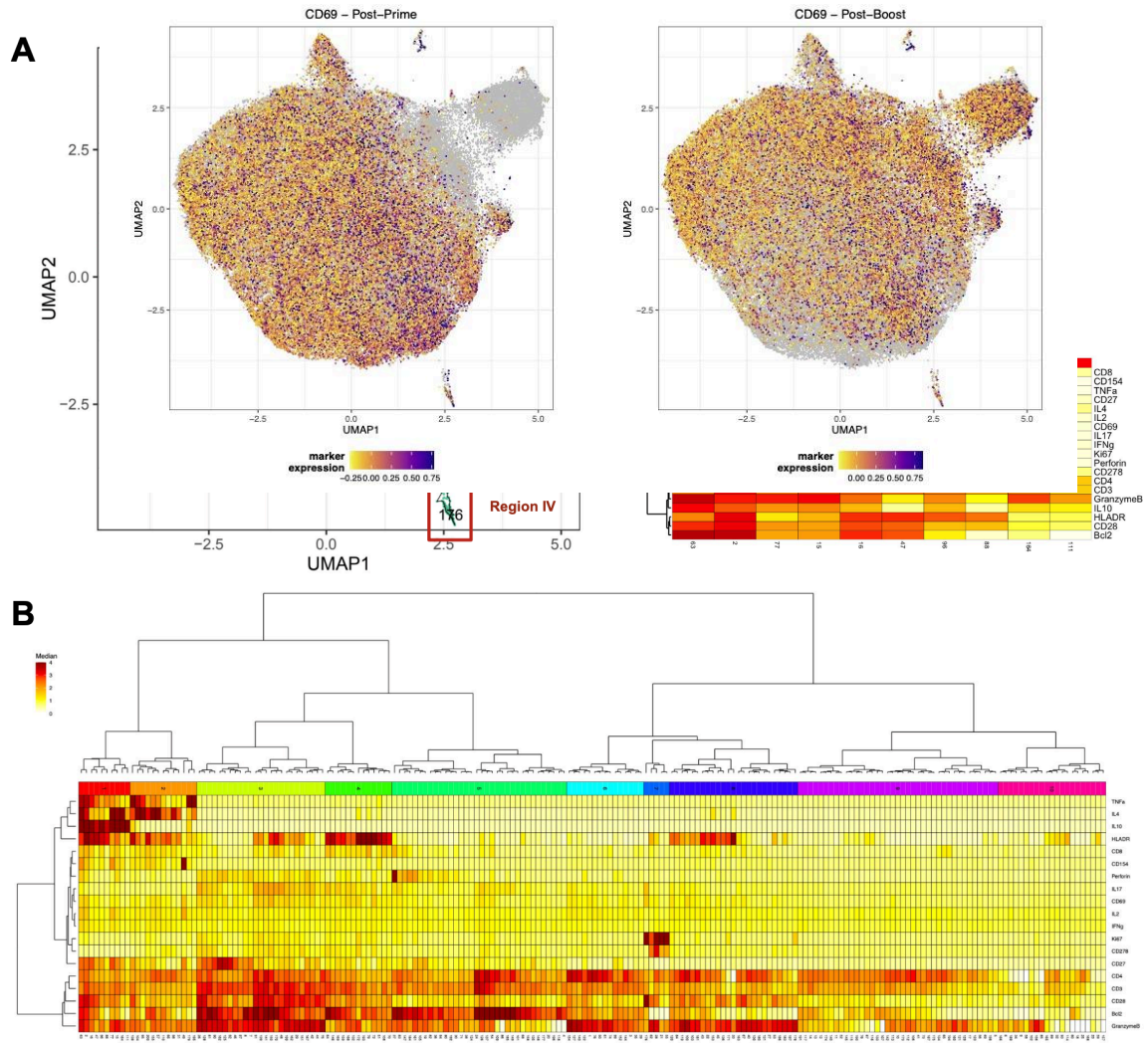


Figure 4. UMAP-based identification of distinct T-cell regions and their phenotypic characterization (A) UMAP visualization of single T cells with k-means clustering ($k = 200$) performed in manifold space. Individual clusters are delineated and annotated, and four spatially distinct regions (Regions I–IV) are highlighted based on their separation from the central phenotypic manifold. (B) Heatmap showing median expression of lineage, activation, co-stimulatory, proliferation, cytokine, and cytotoxic markers across all clusters. Hierarchical clustering of clusters based on marker expression reveals phenotypically coherent groups that correspond to the UMAP-defined regions, supporting the biological relevance of spatially segregated clusters. (C) Heatmap of Region III megacluster following cluster merging. Clusters comprising Region III (clusters 63, 2, 16, 47, 96, 15, 77, 88, 111, and 164) were merged into a single metacluster based on shared expression profiles.

II - Longitudinal dynamics and phenotypic evolution of T-cell populations following prime-boost immunization

Global evolution of the T-cell populations

Visual observations of activation marker overlays reveal a marked intensification of the immune landscape following the booster immunization, signifying a transition from a broad primary response to a highly specialized secondary response (**Fig. 4A**).

A.

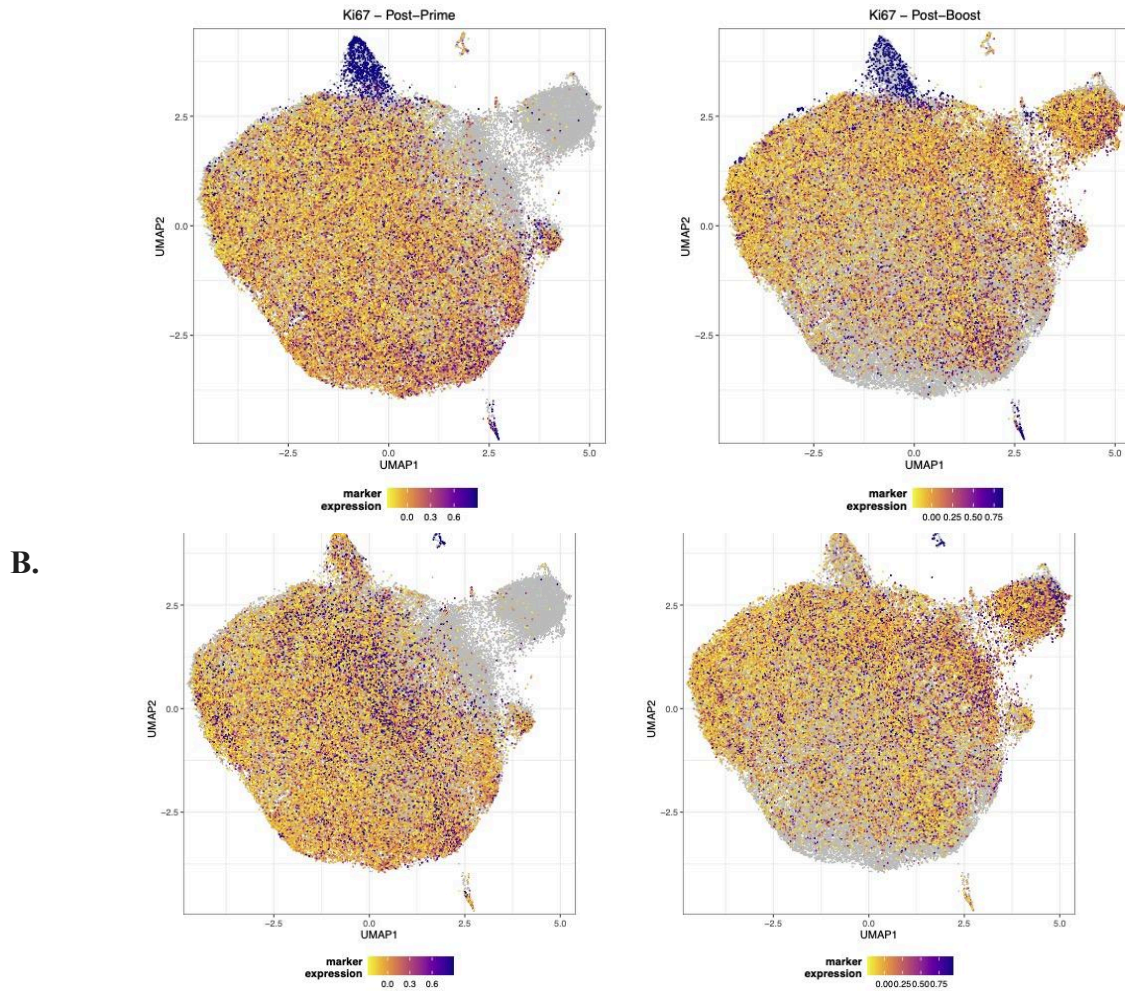


Figure 5. Spatiotemporal evolution of functional activation markers expression across the T-cell manifold. (A) Longitudinal expression of CD69 post-prime and post boost. **(B)** Longitudinal expression of CD154 post-prime and post boost.

The expression of CD69 (**Fig. 5A**) and CD154 (**Fig. 5B**) exhibits a robust increase in **Region II** and **Region III** post-boost compared to the post-prime stage. As the earliest activation marker newly synthesized and expressed during T lymphocyte activation, the strong presence of CD69 in these regions suggests a rapid and effective response to the MVA-based stimulus. This successful induction is of particular note given that CD69 expression is often severely compromised in advanced clinical stages (11). The high-intensity expression observed here indicates that the prophylactic vaccine model retains the capacity for early-stage T-cell signaling that is otherwise lost during disease progression.

Conversely to this early signaling, the upregulation of CD154 provides evidence of deeper functional maturation. Established as a reliable marker for identifying antigen-specific CD4⁺ T cells, CD154 is induced specifically upon cognate antigen recognition (12). Its concentration within these peripheral regions identifies them as the primary functional sites of vaccine-induced helper activity. This observation contrasts with immunocompromised states where lower CD154 expression is usually associated with a failure to develop seroprotective antibody levels (13).

A.

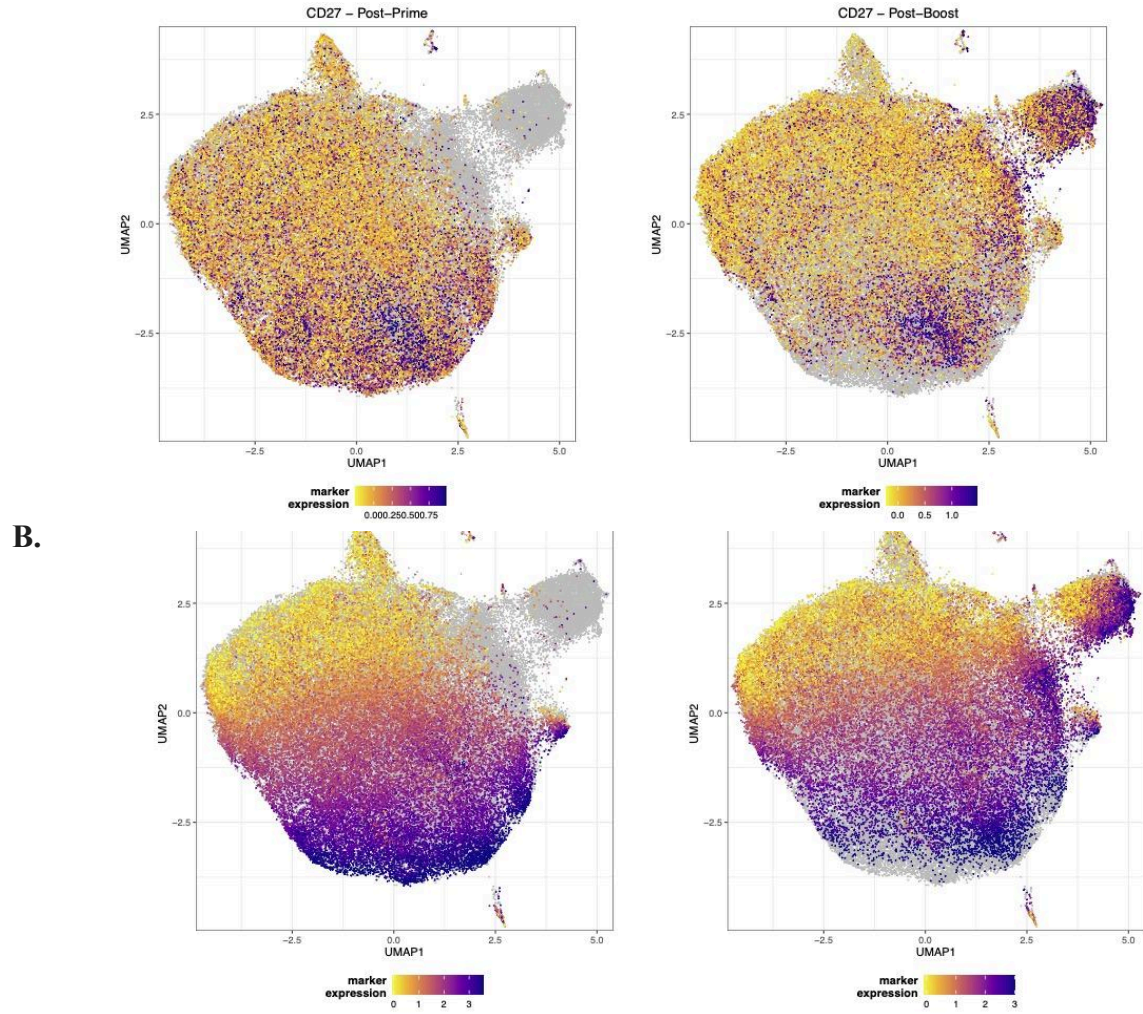
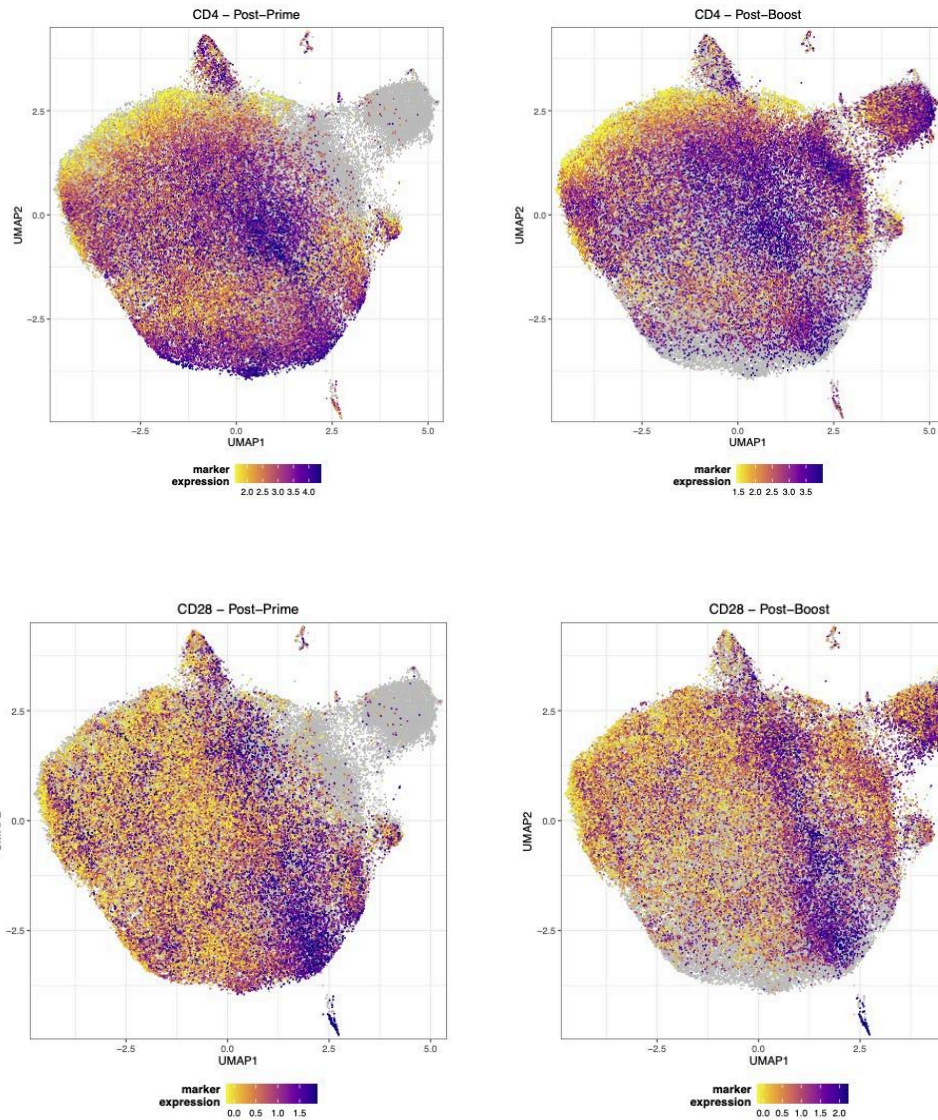


Figure 6. Spatiotemporal evolution of functional proliferation markers expression across the T-cell manifold. (A) Longitudinal expression of Ki67 post-prime and post boost. (B) Longitudinal expression of Bcl2 post-prime and post boost.

In this model, the co-localization of high-intensity CD154 and the proliferation marker Ki67 (**Fig 6A**) within **region III** post-boost provides a clear functional fingerprint. As noted by previous studies, these CD4+Ki67+ populations likely represent circulating effector cells involved in immune surveillance, proving that the booster dose successfully recruits a specialized fraction of the T-cell pool into a recently activated, functionally licensed effector state (14). Additionally, Bcl-2 expression maps (**Fig. 6B**) reveal a significant intensification in **region III** following the booster immunization. Biologically, this increase in Bcl-2 is a key marker of a stabilized immune response as it is essential for the persistence of memory populations. In HIV research, high Bcl-2 levels are known to protect infected cells within the latent reservoir, allowing them to resist being killed by the immune system's granzyme and perforin pathways. In the context of this vaccine model, the enrichment of Bcl-2 in Region III suggests that the booster has successfully transitioned activated effector cells into a durable, long-lived memory population capable of persisting over time (15).

A.



B.

Figure 7. Spatiotemporal evolution of costimulation markers expression across the T-cell manifold. (A) Longitudinal expression of CD27 post-prime and post boost. **(B)** Longitudinal expression of CD28 post-prime and post boost.

The transition from quiescent to effector states is further corroborated by the expression patterns of costimulation and survival markers. The co-stimulatory receptors CD27 (**Fig. 7A**) and CD28 (**Fig. 7B**) demonstrate high intensity within the central, dense manifold at baseline and post-prime, consistent with a naive or central-memory differentiation state. Conversely, these markers show diminished intensity in the peripheral effector "islands" while concurrently increasing within **Region III** and **Region IV** following the booster dose. This pattern visually confirms the migration of vaccine-specific cells out of the resting central manifold and their maturation into an effector phenotype.

A.

B.

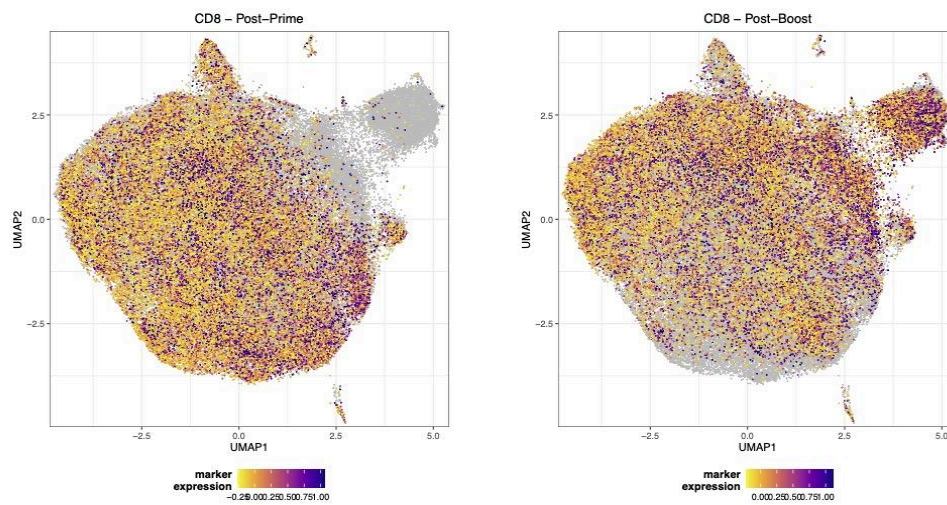


Figure 8. Spatiotemporal evolution of lineage markers expression across the T-cell manifold. (A) Longitudinal expression of CD4 post-prime and post boost. **(B)** Longitudinal expression of CD8 post-prime and post boost.

The longitudinal manifold shifts are characterized by distinct spatial dynamics between the CD4⁺ and CD8⁺ lineages. Post-boost overlays demonstrate an expansion of CD4 expression (**Fig. 8A**) in Region III and the southern portion of the central manifold. In contrast, CD8 expression (**Fig. 8B**) intensity increases significantly within **region III** following the booster dose but shows a concomitant decrease in the southern central zone. This spatial partitioning reflects a transition from the broad proliferative response seen post-prime and driven by the Ki67⁺ proliferative zone in **Region I** (clusters 21, 55, 28), to a targeted, specialized secondary response. The booster matures this landscape by recruiting antigen-responsive helper T cells in **Region II** (cluster 49), uniquely identified by high CD154 expression, which are critical for coordinating *de novo* adaptive immunity. These kinetics culminate in the expansion of the **Region III** metacluster (clusters 63, 111, 164), which exhibits a survival phenotype. By coordinating Bcl-2 and CD28 upregulation with the production of IL-10, IL-4, and granzyme B, this CD4-enriched population represents a successful transition from transient activation to a multifunctional memory state. On the other side, **Region IV** (cluster 176) sustains a proliferating pool that retains co-stimulatory capacity over time.

In the context of our HIV vaccine, these lineage-specific shifts demonstrate that the prime-boost regimen orchestrates a mature immune landscape, balancing immediate

cytotoxic effector potential with a survival-prone helper network essential for long-term immune surveillance.

Quantitative identification of vaccine-responsive clusters

1. Peak effector response vs. Baseline

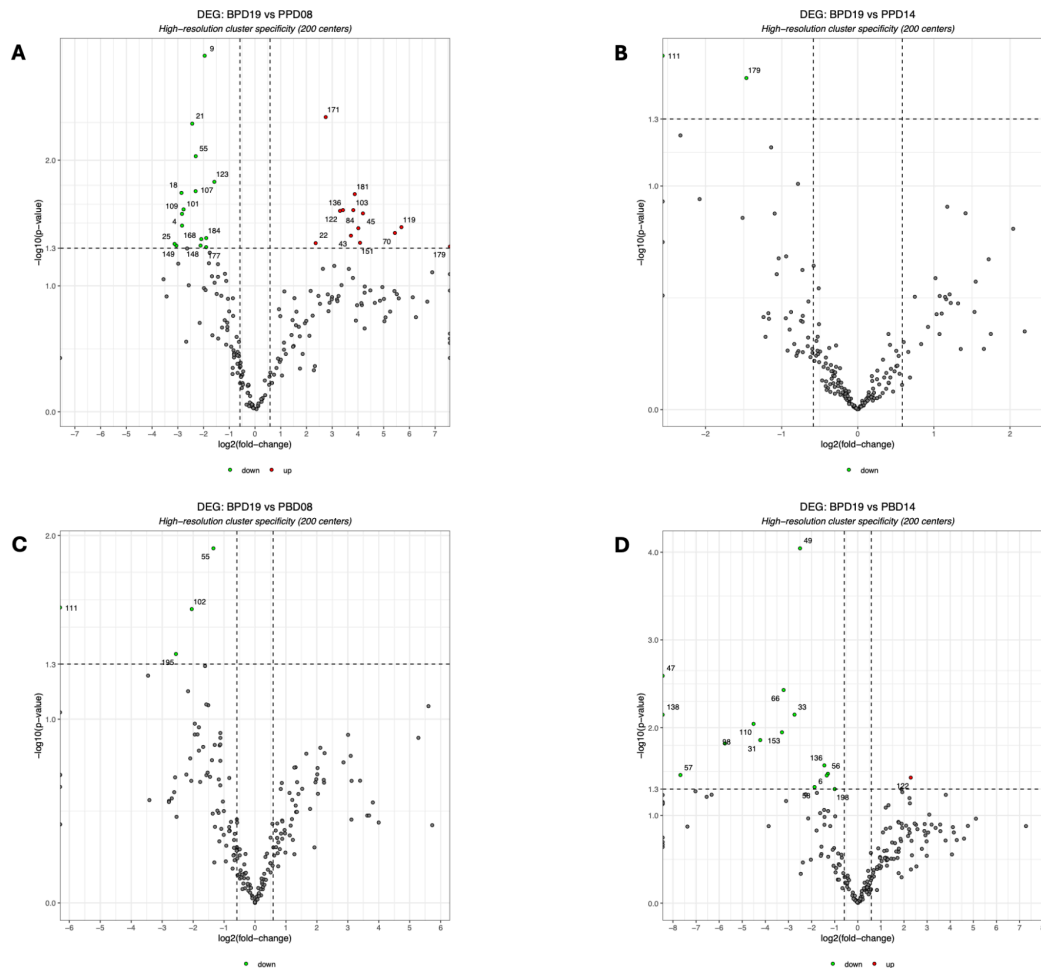


Figure 9. Statistical identification of vaccine-responsive clusters relative to baseline. (A) Volcano plot comparing cluster abundances at peak prime (PPD08) versus baseline (BPD19). (B) Volcano plot comparing cluster abundances at peak prime (PPD14) versus baseline (BPD19). (C) Volcano plot comparing cluster abundances at peak boost (PPD08) versus baseline (BPD19). (D) Volcano plot comparing cluster abundances at peak boost (PPD14) versus baseline (BPD19).

The quantitative impact of the MVA-based HIV vaccine is characterized by a significant redistribution of T-cell abundances across the phenotypic manifold, with the most dramatic shifts occurring during the peak effector phases (Day 8) of both the prime and boost. Statistical analysis via volcano plots reveals that the post-prime response (**Fig. 9A**) is defined by a robust expansion of clusters within the effector regions, notably cluster 171 and cluster 181. These clusters exhibit a significant positive fold-change relative to baseline, identifying them as the primary "first responders" to the initial antigen encounter. Interestingly, while **region I** is the proliferative zone of the response, clusters 21 and 55 appear significantly reduced in abundance at PPD08 compared to baseline. In high-dimensional space, this

quantitative decrease suggests a rapid phenotypic exit, where cells initially occupying these cycling clusters at baseline have differentiated and migrated into more mature effector clusters by day 8. As the response progresses to 14 days post prime, the first dose vaccine response remains characterized by a higher abundance of early effector and proliferative clusters, such as clusters 28 and 176, compared to the secondary response. By day 14 (**Fig. 9B**), the prime-induced response demonstrates a rapid return toward baseline equilibrium, primarily characterized by the significant contraction of Cluster 111 and Cluster 179 relative to pre-vaccination levels.

Following the booster immunization, the immune landscape undergoes a secondary, more specialized quantitative transition. 8 days after the boost (**Fig. 9C**), significant abundance decreases are observed in clusters 55, 102, and 111 relative to baseline. This trend intensifies 6 days later (**Fig. 9D**), where cluster 49, the **region II** population uniquely identified by its high CD154 expression, shows the most statistically significant reduction in abundance. Biologically, this reflects the completion of the "recall" phase; the rapid recruitment of functional helper T cells is transient, and by PBD14, these cells have either contracted or transitioned into the quiescent memory clusters established in the central manifold.

2. Post-Boost vs. Post-Prime

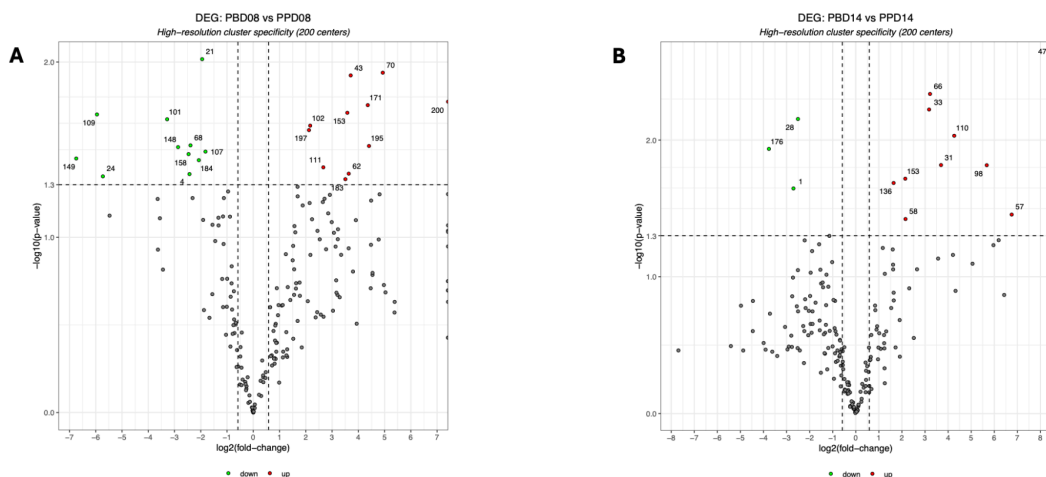


Figure 10. Statistical identification of vaccine-responsive clusters at peak post boost relative to peak post-prime. (A) Volcano plot comparing cluster abundances at peak boost (PBD08) versus peak prime (PPD08). **(B)** Volcano plot comparing cluster abundances at peak boost (PBD14) versus peak prime (PPD14).

Statistical comparison of the peak and resolution phases reveals that the booster dose fundamentally matures the T-cell landscape rather than merely replicating the primary response. At the peak of expansion (**Fig. 10A**), a "booster effect" is observed, characterized by the significant amplification of mature effector populations. Clusters 43, 70, 153, and 171 exhibit both higher statistical significance and greater fold-change magnitude following the second dose, identifying them as the primary drivers of the recall response. Conversely, the significant reduction in the abundance of cluster 21, described as a central proliferative hub in **region I**, suggests that the booster response is kinetically accelerated. In this secondary phase, cells appear to exit the initial cycling state more rapidly to occupy specialized effector niches, indicating a more efficient transition from proliferation to functional maturation.

By the resolution phase (**Fig. 10B**), quantitative analysis highlights a qualitative refinement of the immune memory pool. Cluster 47 demonstrates a massive expansion ($\log_2$ fold-change ≈ 8) post-boost, alongside significant increases in clusters 66, 33, 110, and 31, which are associated with specialized cytokine production and functional longevity. This expansion is contrasted by a significant decrease in proliferative and early effector clusters, such as clusters 28 and 176, relative to their post-prime levels.

3. Memory/Contraction phase vs. Peak response

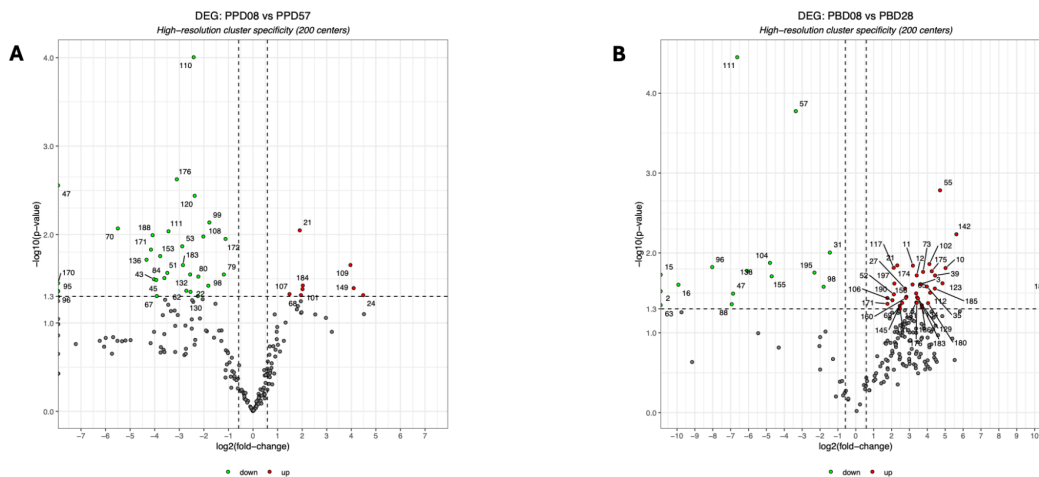


Figure 11. Statistical identification of vaccine-responsive clusters at peak post boost relative to peak post-prime. (A) Volcano plot comparing cluster abundances at peak prime (PPD08) versus late prime (PPD57). **(B)** Volcano plot comparing cluster abundances at peak boost (PBD08) versus late boost (PBD28).

The transition from peak effector expansion to long-term immune stabilization is defined by a systematic filtering of the T-cell pool, where transient proliferative signals are cleared in favor of a durable memory footprint.

Following the primary immunization (**Fig. 11A**), we can observe that the central proliferative hub, cluster 21, undergoes a marked reduction from peak levels, as evidenced by its significant positive fold-change relative to the day 57 stabilization point. In contrast, clusters 110 and 176 are significantly enriched in the negative fold-change region, identifying them as key components of the late-stage manifold and the successful establishment of a survival-oriented "survival" footprint that persists for nearly two months post-prime. Statistical analysis comparing the peak boost response to the final memory phase (**Fig. 11B**) identifies a significant contraction of the **region I** proliferative mechanism, notably clusters 55 and 142, which exhibit strong positive fold-changes, confirming their dominance during the acute expansion phase and their subsequent resolution by day 28. This contraction is balanced by the persistence of the **region III** effector-memory metacluster; clusters such as 111 and 57 show negative fold-changes in this comparison, indicating they probably characterize the mature, stabilized landscape rather than the early recall flare.

Collectively, these shifts demonstrate that this prime-boost regimen effectively develops the T-cell manifold, balancing a transitory cytotoxic burst with the consolidation of a specialized memory reservoir needed for long-term immune surveillance.

Kinetic Evolution and Functional Maintenance of Populations

To systematically characterize the temporal dynamics of vaccine-responsive T-cell populations, we performed longitudinal abundance profiling across all major experimental timepoints spanning from baseline (BPD19) through the primary response (PPD03, PPD08, PPD14, PPD57) to the booster recall phase (PBD03, PBD08, PBD14, PBD28). By tracking cluster-specific abundance patterns via boxplot visualization, we classified vaccine-responsive populations into three distinct kinetic waves based on their temporal expansion and contraction profiles: **early transient responders**, **effector peak populations**, and **stabilized memory clusters**. This stratification reveals not only the temporal coordination of the immune response but also the phenotypic plasticity underlying the transition from acute activation to durable immunological memory.

Early Transient Responders: Rapid Activation and Phenotypic Exit

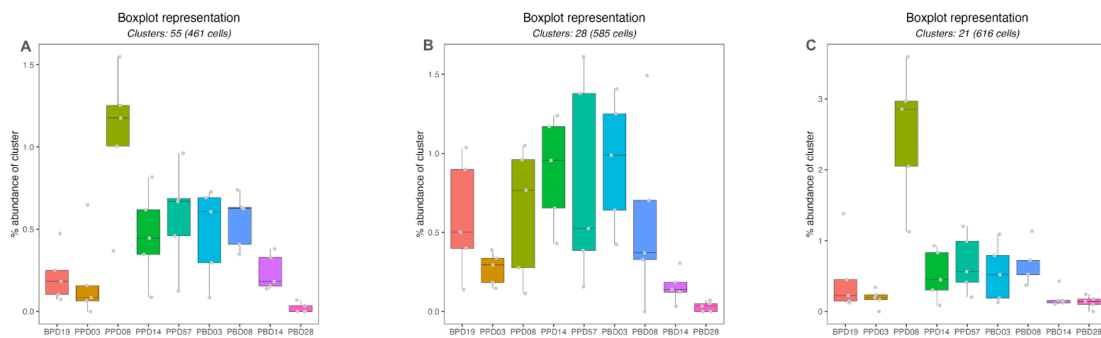


Figure 12. Early transient responder kinetics in Region I proliferative clusters. Boxplots show longitudinal abundance of clusters 55 (A), 28 (B), and 21 (C) across baseline, post-prime, and post-boost timepoints.

The first kinetic wave is characterized by populations that undergo rapid, transient expansion within the first 3–8 days following immunization, followed by swift contraction as the response matures. **Cluster 21** (Fig. 12C) located within **Region I**, consistent with its significant enrichment at PPD08 relative to PPD57 in the PPD08 vs PPD57 differential abundance analysis (Fig. 11A), displaying striking abundance peaks at day 8 post-prime (PPD08) with median values approaching ~3% of the cluster composition. This cluster is phenotypically defined by high Ki67 expression with minimal cytotoxic markers, identifying it as the central proliferative hub. By day 14 post-prime (PPD14), cluster 21 contracts sharply toward baseline and remains suppressed through the boost phase, indicating its role is restricted to de novo proliferation during primary antigen encounter.

Cluster 55, (Fig. 12A), also located within **Region I**, follows an identical early-transient trajectory, peaking at PPD08 before declining to baseline by PPD14 and remaining suppressed throughout the boost phase. Phenotypically, it is elevated in Ki67 alongside several activation-associated markers, confirming an activated proliferative state. **Cluster 28** (Fig. 12B) in **Region I**, occupies an intermediate position within the early transient wave. This cluster exhibits moderate expansion at PPD08 with sustained presence through PPD14. Uniquely, cluster 28 combines high Ki67 with elevated CD278 (ICOS) and moderate Granzyme B, indicating a transitional state bridging proliferation and effector

differentiation. The temporal persistence of cluster 28 relative to clusters 21 and 55 suggests it represents a bridge between pure proliferation and functional maturation. Following the boost, all three clusters remain suppressed, demonstrating the booster bypasses initial proliferative checkpoints through accelerated memory-driven kinetics.

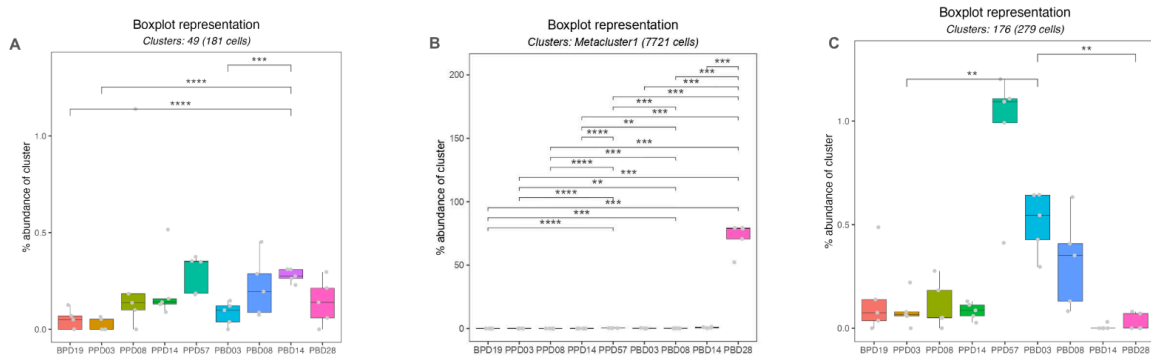


Figure 13. Effector activation, memory stabilization, and late proliferative kinetics. Boxplots show longitudinal abundance of cluster 49 (A), Region III metacluster (B), and cluster 176 (C) across baseline, post-prime, and post-boost timepoints.

Effector Peak Populations: Sustained Functional Activation

Cluster 49 (Fig. 13A), the defining population of **Region II**, exhibits a specialized biphasic kinetic profile consistent with antigen-specific helper T-cell activation. This cluster is characterized by high CD154 (CD40L) expression, a canonical marker of antigen-engaged CD4⁺ T cells required for co-stimulatory signaling and immune coordination. Longitudinal abundance analysis shows modest expansion during the primary response, followed by a pronounced but transient increase following booster immunization (PBD03–PBD08), with rapid contraction by PBD14. This transient post-boost expansion is consistent with rapid mobilization of antigen-experienced helper T cells during recall responses.

The contraction following peak expansion, together with the absence of sustained persistence, indicates that cluster 49 represents a transient effector population involved in functional activation rather than long-term memory maintenance. These kinetics support a model in which antigen-experienced helper T cells are rapidly recruited upon booster immunization to deliver effector function before contracting as stabilized memory populations emerge.

Stabilized Memory Clusters: Long-Term Functional Maintenance

The third kinetic wave comprises populations that undergo progressive accumulation and persist at stable, elevated abundance from the late prime phase through the post-boost memory phase, representing successful establishment of durable immunological memory. These populations are phenotypically distinguished by coordinated Bcl-2 and CD28 upregulation, indicating survival signaling and co-stimulatory competence essential for long-lived memory T cells.

Region III Metacluster: The most striking feature of the stabilized memory wave is the coordinated behavior of the **Region III metacluster** (Fig. 13B) (clusters 63, 2, 16, 47,

96, 15, 77, 88, 111, 164). This megacluster undergoes progressive, highly significant accumulation beginning at PPD08, continuing through PPD57, and further expanding post-boost to peak at PBD08 ($p < 0.0001$ across multiple comparisons). Critically, the metacluster maintains elevated abundance through PBD28, demonstrating stabilization at a new steady-state that persists three months post-prime rather than undergoing transient effector contraction.

Phenotypically, Region III is uniquely defined by coordinated **Bcl-2 and CD28 upregulation**, canonical long-lived memory markers providing apoptosis protection and co-stimulatory competence. Additional **IL-10, IL-4, and Granzyme B** expression indicates multifunctional, polyfunctional capacity balancing regulatory cytokines with cytotoxic potential. This combination defines the Region III metacluster as a multifunctional, polyfunctional memory population capable of coordinating helper, regulatory, and cytotoxic responses. **CD4 enrichment** combined with post-boost expansion kinetics identifies this as the primary vaccine-induced CD4⁺ memory compartment in this MVA-based HIV vaccine model. Individual clusters within the Region III metacluster, exhibited progressive enrichment and sustained persistence during late post-boost timepoints, consistent with stabilization into survival-associated memory populations (Supplementary Fig. 3).

Cluster 176, (Fig. 13C) the sole constituent of **Region IV**, exhibits a delayed and biphasic kinetic profile distinct from other populations. Abundance remains near baseline during early post-prime timepoints, followed by significant expansion peaking at PPD57 (median ~1.1%, $p < 0.01$), consistent with delayed recruitment during the late primary response (Fig. 11A). Following booster immunization, cluster 176 undergoes marked contraction, declining to near-baseline levels by PBD14 with only modest recovery at PBD28 (Fig. 10B). Phenotypically characterized by concurrent CD28 and Ki67 expression, cluster 176 represents a proliferative population retaining co-stimulatory capacity. Unlike Region III memory-associated populations, which expand and stabilize following booster immunization, the contraction of cluster 176 suggests it represents a transient proliferative population associated primarily with the primary response rather than a durable memory population. The failure of this population to persist or re-expand during the memory phase indicates that it likely represents a transitional proliferative state that is progressively replaced by more stable memory populations following booster immunization.

Discussion

Discussion I. Yaroslava Shevchenko

In this study, we applied high-dimensional single-cell analysis to characterize the dynamics of vaccine-induced T-cell responses following prime–boost immunization with an MVA-based HIV vaccine. Using mass cytometry combined with unsupervised dimensionality reduction and clustering, we resolved phenotypically distinct T-cell populations and tracked their evolution across multiple time points. This approach revealed a complex immune landscape comprising proliferative subsets, activation-associated populations, and a prominent vaccine-responsive metacluster that expanded following booster immunization. Together, these findings provide a detailed single-cell view of how prime–boost vaccination shapes T-cell phenotypes over time.

To capture the full breadth of vaccine-induced heterogeneity, we used high-resolution clustering of the UMAP manifold ($k = 200$). This overclustering strategy was chosen to minimize the risk of missing rare but biologically relevant populations, a known challenge in single-cell immune profiling (16). The resulting manifold structure revealed both a dense central compartment and multiple spatially segregated peripheral regions, each corresponding to distinct and coherent T-cell states. Among these regions, proliferative clusters enriched for Ki67 localized predominantly to discrete peripheral areas, consistent with transient clonal expansion following antigen encounter (7). Cluster 49, defined by high CD154 (CD40L) expression, formed an isolated population, supporting its annotation as an antigen-responsive helper T-cell subset (8). Of particular interest, Region III comprised a group of spatially adjacent clusters with highly overlapping phenotypic profiles that were therefore merged into a single metacluster for downstream analyses. This metacluster displayed coordinated upregulation of CD28 and Bcl-2, along with increased expression of IL-10, IL-4, Granzyme B, and CD4, indicating a CD4-enriched population combining survival capacity, co-stimulatory signaling, cytokine production, and partial effector features. Notably, this metacluster expanded markedly following booster immunization and accounted for approximately 50% of clustered cells at the late post-boost time point (PBD28), highlighting it as a dominant vaccine-responsive population. The phenotype of this metacluster aligns with emerging models of vaccine-induced T-cell immunity, in which durable responses are mediated by multifunctional and phenotypically flexible populations rather than narrowly defined subsets (1). Elevated CD28 and Bcl-2 expression has been associated with enhanced persistence and memory potential, while cytokine production is characteristic of helper and regulatory programs observed following viral vaccination (6). The concurrent expression of Granzyme B within a CD4-skewed population further supports the concept of functional plasticity in vaccine-elicited T cells, as previously reported in both human and non-human primate studies (18). These findings reinforce the idea that effective prime–boost vaccination promotes the emergence of long-lived, multifunctional T-cell states rather than transient effector populations alone.

Several limitations of this study should be considered. From a biological perspective, analyses were restricted to peripheral blood, which may not fully reflect tissue-resident or lymphoid T-cell populations that are critical for antiviral immunity (19). In addition, the relatively small cohort size limits statistical power and may underestimate inter-individual variability. From a computational standpoint, dimensionality reduction techniques such as UMAP inevitably distort global distances and neighborhood relationships, meaning that spatial proximity in the manifold should be interpreted qualitatively rather than quantitatively (17). Clustering outcomes are also sensitive to parameter selection, including the number of clusters, which can influence cluster boundaries and downstream interpretations. Although overclustering helps preserve rare populations, it may also fragment continuous phenotypic gradients, necessitating merging strategies such as metaclustering. Additional methodological limitations include potential batch effects between samples and restriction to a predefined marker panel, which may omit relevant functional pathways. Moreover, phenotypic inference was based on marker expression without direct functional validation, such as antigen-specific stimulation or cytokine secretion assays, limiting causal interpretation of cluster function. These constraints are common to high-dimensional cytometry studies and highlight the importance of integrating complementary approaches.

Future studies could extend this work by incorporating functional assays and T-cell receptor sequencing to directly link phenotype, function, and clonal dynamics. Increasing cohort size and including tissue compartments would improve resolution of vaccine-induced heterogeneity and durability. Methodologically, combining multiple clustering algorithms and benchmarking results across approaches may further strengthen robustness. Finally, comparative analyses across different vaccine platforms could determine whether the emergence of a dominant survival- and cytokine-associated metacluster represents a general hallmark of effective prime–boost immunization strategies.

In conclusion, this study demonstrates that MVA-based prime–boost vaccination induces a heterogeneous and dynamic T-cell response characterized by discrete proliferative populations and a dominant vaccine-responsive metacluster that emerges after boosting. These findings advance our understanding of cellular correlates of vaccine-induced immunity and offer insights relevant to the rational design and evaluation of next-generation HIV vaccines and other T-cell–targeted immunization strategies.

Discussion II. Margaux Beze

This study provided a high-dimensional, longitudinal map of the T-cell response to an MVA-based HIV vaccine in a non-human primate model. By integrating single-cell mass cytometry (CyTOF) with unsupervised clustering and statistical differential abundance analysis, we characterized the immune environment across the transition from primary activation to the establishment of stable memory. The results demonstrated that the prime-boost regimen does not simply amplify existing populations but fundamentally restructures the T-cell populations to favor a refined, survival-oriented immune reservoir (4).

The phenotypic diversity of the T-cells was found to be induced by the vaccine, with the lymphocytes populations being segregated into four distinct functional regions. The initial response is characterized by Region I, defined by high Ki67 expression and proliferative attributes. This then transitions into a specialized antigen-responsive helper niche in Region II, uniquely marked by the transient expression of CD154. The secondary response culminates in the formation of a high-intensity effector-memory metacluster in Region III, which balances cytotoxic potential (with granzyme B) and survival signals (Bcl-2, CD28) (20, 21). This spatial distribution demonstrates that the MVA platform recruits a broad spectrum of T-cell states, ranging from immediate effectors to co-stimulatory helper networks.

We conducted further statistical analysis to address how these populations evolved across time and more specifically how the prime-boost regiment affected them. We confirmed that they progressed through a series of defined phases: early activation and broad clonal expansion, a targeted contraction phase, and a resolution phase with long-lived memory populations. The secondary immunization showed evidence of a "booster effect" and a subsequent filtering mechanism as it triggered a shift, with the massive expansion of effector-memory CD4⁺ T cells (cluster 47) compared to the primary response (22). While the peak of both the prime and boost is dominated by effector expansion (clusters 171, 181), the system systematically cut back transitory activation clusters such as the CD154-high (cluster 49) population during the resolution phase. This resolution favored the stabilization of "survivor" clusters (clusters 184, 169, 150), which maintained a high prevalence into the late memory phase. This ensured the immune system returned to homeostasis while still holding on to a high-quality, long-lived immune surveillance from the lymphocytes.

While high-dimensional phenotypic mapping provides unparalleled resolution of the circulating T-cell landscape, several constraints limit the direct translation of these findings to protective efficacy. Among these is the reliance on peripheral blood mononuclear cells (PBMCs) as a proxy for systemic immunity. Given that HIV predominantly replicates and establishes reservoirs within lymphoid and mucosal tissues, the current focus of this study being the peripheral compartment may not fully capture the dynamics of tissue-resident memory populations, which constitute the critical first line of defense at sites of viral entry. Furthermore, the absence of integrated T-cell receptor sequencing or direct MHC-multimer staining prevents the establishment of a correlation between the observed cluster expansions

and the specific HIV-antigen recognition. Consequently, it remains unclear whether these specialized phenotypes represent a broad response to the MVA vector itself or a targeted enrichment of HIV-specific effector-memory cells. Finally, while the non-human primate model is a rigorous surrogate for human immunogenicity, the degree of cross-species conservation regarding these specific cluster signatures and their respective protective thresholds requires further clinical validation.

Looking forward, the identification of a stable effector memory footprint, particularly within the Region III metacluster, entails long-term longitudinal studies to determine the ultimate half-life and functional exhaustion profile of these populations beyond the 2-months observation window. Future investigations should prioritize the integration of these high-dimensional phenotypic maps with potent viral challenge models to establish whether the quantitative expansion of multifunctional subsets serves as a true correlate of protection against HIV or as a viral set-point control. Refining the ability to steer the T-cell response from a broad and transient inflammatory flare toward an efficient, survival-oriented reservoir will be essential for the design of the next generation of prophylactic HIV vaccines.

In conclusion, this study shows that the MVA-based HIV vaccine reshapes the T-cell response in a coordinated and time-dependent manner. The prime–boost strategy does more than just increase cell numbers; it guides the immune response from early activation and strong proliferation toward a more stable and long-lasting memory state. By using high-dimensional CyTOF analysis with unsupervised clustering and longitudinal statistics, we managed to track this progression from expansion to phase showing the persistence of selected effector-memory populations. In particular, the sustained presence of Region III cells highlights the vaccine’s ability to generate T cells that combine functional strength with survival capacity. Although additional studies in tissues, with antigen-specific tracking and viral challenge models, are needed to confirm their protective role, these results provide a clearer picture of how vaccine-induced T-cell responses mature over time. Overall, effective HIV vaccine design will require strong initial responses coupled with durable, efficient memory T cells in order to obtain a robust long-term immune surveillance.

Discussion III . Shubhdeep Kaur

This study provides a high-dimensional, longitudinal characterization of T-cell responses induced by an MVA-based HIV vaccine in a prime–boost macaque model. By integrating UMAP-based visualization, unsupervised clustering, and temporal abundance tracking, we identified coordinated kinetic waves corresponding to early proliferation, transient effector activation, and stabilized memory formation. Collectively, the data support a model in which booster immunization preferentially amplifies antigen-experienced memory populations rather than reinitiating broad de novo proliferation, consistent with established paradigms of T-cell differentiation and recall responses (23,24).

Longitudinal analysis revealed three sequential kinetic phases. Early transient responder clusters (21, 55, 28) were characterized by elevated Ki67 expression with limited effector marker expression, consistent with proliferative precursor states. Ki67 is expressed during active phases of the cell cycle but is absent in quiescent cells (25), and has been widely used to quantify antigen-specific lymphoproliferation (26). These clusters peaked at day 8 post-prime and contracted by day 14, aligning with classical expansion–contraction kinetics of vaccine-induced T-cell responses (23,24). Their contraction coincided with expansion of more functionally mature populations, suggesting differentiation from cycling precursors into antigen-experienced effector states.

Recall effector activation was exemplified by cluster 49, which exhibited transient CD154 upregulation following booster immunization. CD154 (CD40L) is rapidly induced upon TCR engagement and identifies antigen-specific CD4⁺ T cells with high sensitivity and specificity (27–29). The short-lived expansion of this cluster during PBD03–PBD08 followed by contraction by PBD14 is consistent with rapid helper mobilization during secondary responses. Importantly, the transient nature of CD154 expression distinguishes acute effector function such as dendritic cell licensing and B-cell help from long-term memory persistence.

Stabilized memory formation was centered on the progressive accumulation of the Region III metacluster from post-prime through late post-boost timepoints, with sustained elevation at PBD28. These cells demonstrated coordinated upregulation of Bcl-2 and CD28 alongside expression of IL-10, IL-4, and Granzyme B. Bcl-2 is essential for memory T-cell survival and resistance to apoptosis (30–32), while CD28 signaling supports survival and optimal recall responses (33). The combined upregulation of Bcl-2 and CD28, together with cytokine production and cytotoxic potential, defines a multifunctional memory-associated phenotype. The presence of Granzyme B within a CD4-enriched population is compatible with emerging recognition of cytotoxic CD4⁺ T-cell programs in vaccination contexts (34). Critically, booster immunization preferentially expanded Region III populations ($p < 0.0001$), whereas early proliferative clusters (21, 55, 28) and late primary cluster 176 did not show sustained expansion. This shift demonstrates that the secondary response is driven primarily by antigen-experienced memory cells rather than renewed naïve proliferation (23,24).

A key limitation of this study is that several markers were excluded during preprocessing due to quality control filtering; important memory markers such as CD45RA, CD197, CD62L,

and CD127 were not included in the final clustering analysis. Therefore, we cannot formally distinguish central memory from effector memory subsets or directly assess lymph node homing and long-term survival programming (36–37).

Other excluded markers, including CD279, CD107a, CD185, FoxP3, CD95, and MIP1 β , limit our ability to evaluate exhaustion, degranulation capacity, Tfh-like differentiation, regulatory T-cell contributions, and detailed trafficking potential. As a result, the classification of Region III as a memory population is based on indirect evidence such as Bcl-2 upregulation, sustained CD28 expression, box plot analysis and long-term persistence rather than formal memory subset definition. Although these features strongly support a memory-associated identity, they do not allow definitive classification.

Additional limitations include the small sample size ($n = 5$ macaques), use of unpaired t-tests without modeling repeated measures, and down-sampling to 25,000 cells, which may reduce detection of rare populations. Clustering was performed in UMAP space rather than in the original high-dimensional marker space (38), which may affect cluster boundaries. Functional validation with antigen-specific restimulation assays was not performed, and only peripheral blood was analyzed, excluding tissue-resident memory compartments that are highly relevant for HIV immunity (39). No trajectory analysis was conducted to formally reconstruct differentiation pathways between proliferative and memory states.

Future studies could improve the analysis by applying multiple testing correction, measuring cytokine co-expression to quantify polyfunctionality, and performing trajectory inference analysis. Alternative high-dimensional cytometry pipelines such as FlowSOM or SPADE could provide complementary insights. These methods are designed to better capture complex phenotypic relationships and rare populations and may improve identification of transitional or memory precursor states. Using multiple clustering approaches would increase analytical robustness and reduce dependence on a single method. Integration of single-cell RNA sequencing and T-cell receptor sequencing could further clarify transcriptional memory programs and clonal dynamics.

In summary, these findings support established models in which primary vaccination induces broad proliferation followed by contraction, while booster immunization promotes a faster, memory-driven recall response (23,24). The preferential expansion of survival-associated, multifunctional CD4⁺ populations after boosting suggests that durable helper memory may be more important than short-term proliferative magnitude for effective HIV vaccine design. Although we could not formally classify central versus effector memory subsets, the Bcl-2⁺CD28⁺ phenotype combined with cytokine production and cytotoxic potential is consistent with long-lived, multifunctional memory T cells (35,40). Focusing on survival-associated and polyfunctional features, rather than response magnitude alone, may improve identification of vaccine strategies capable of inducing durable protective immunity.

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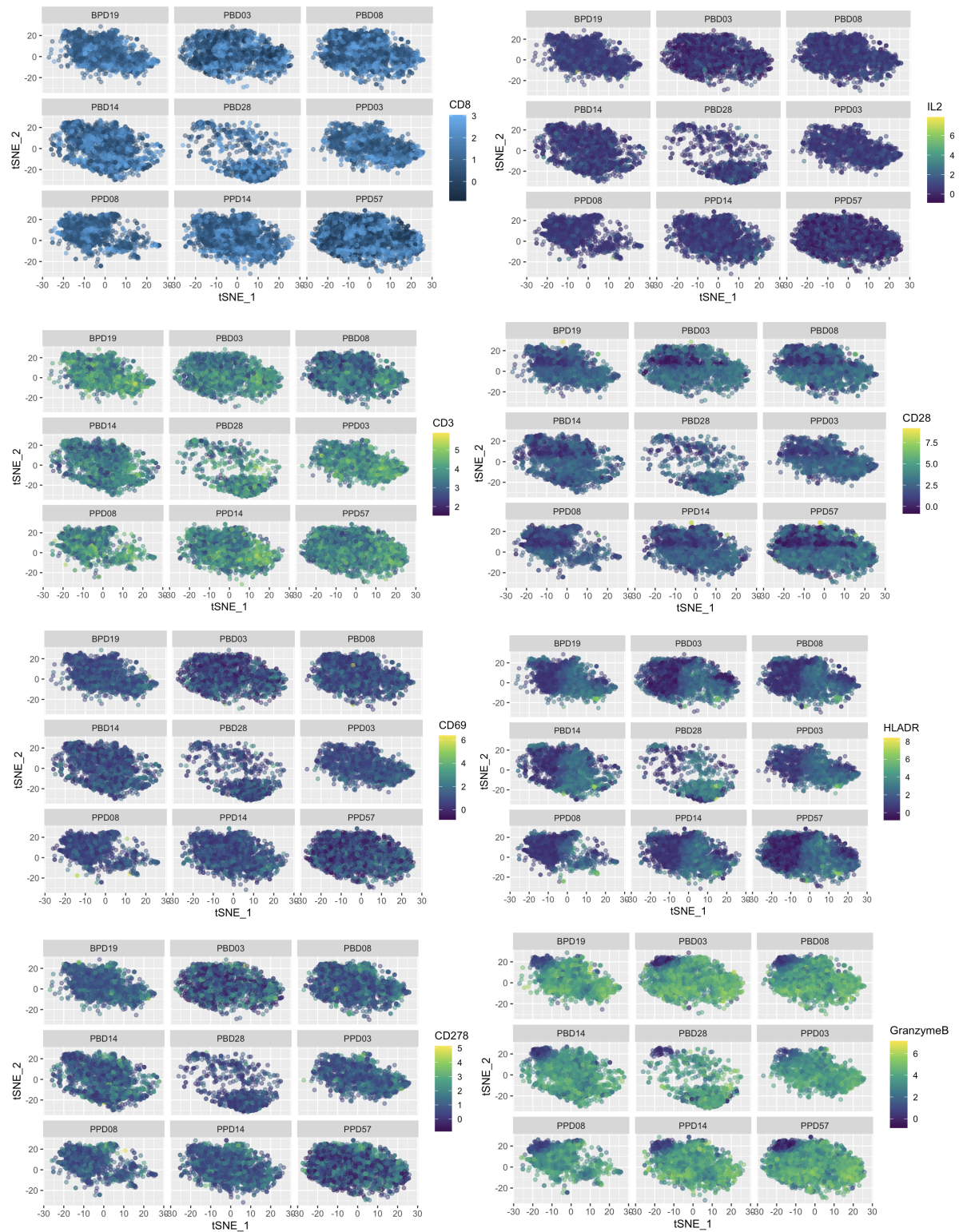
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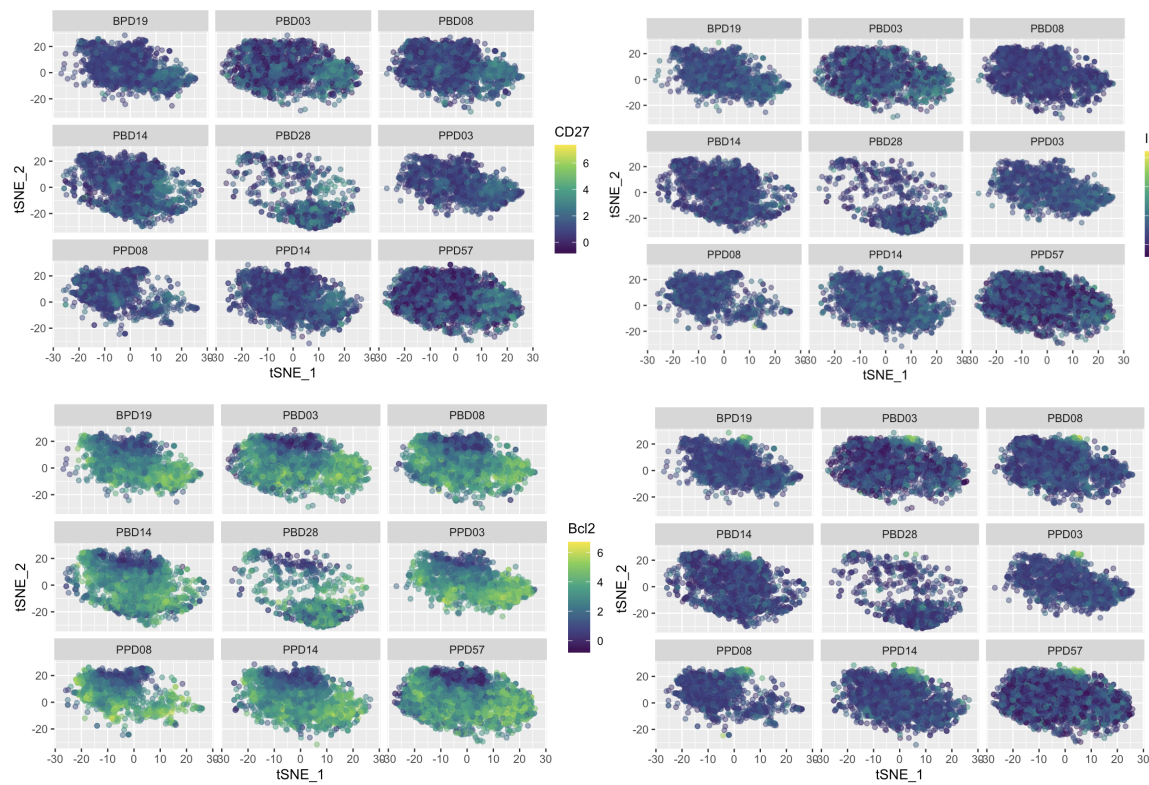
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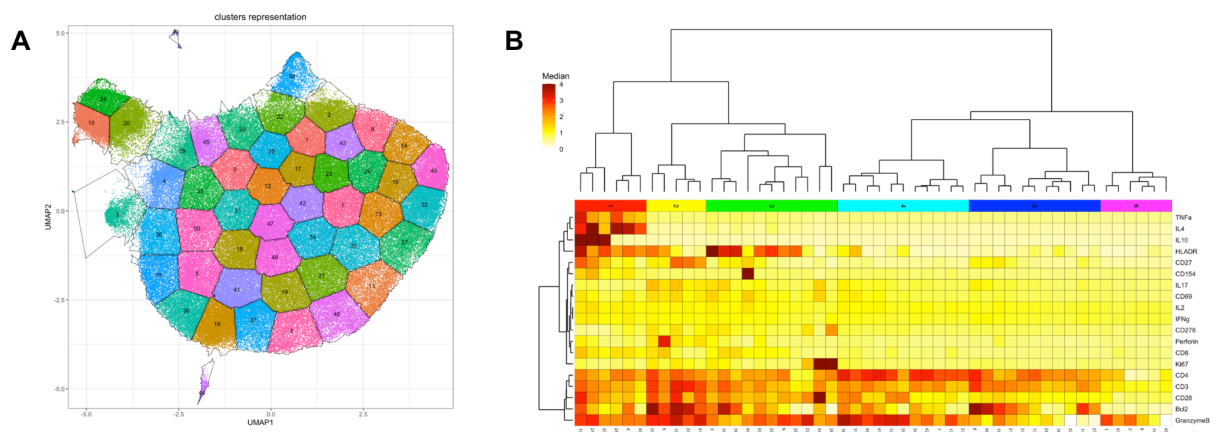
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Supplementary Data



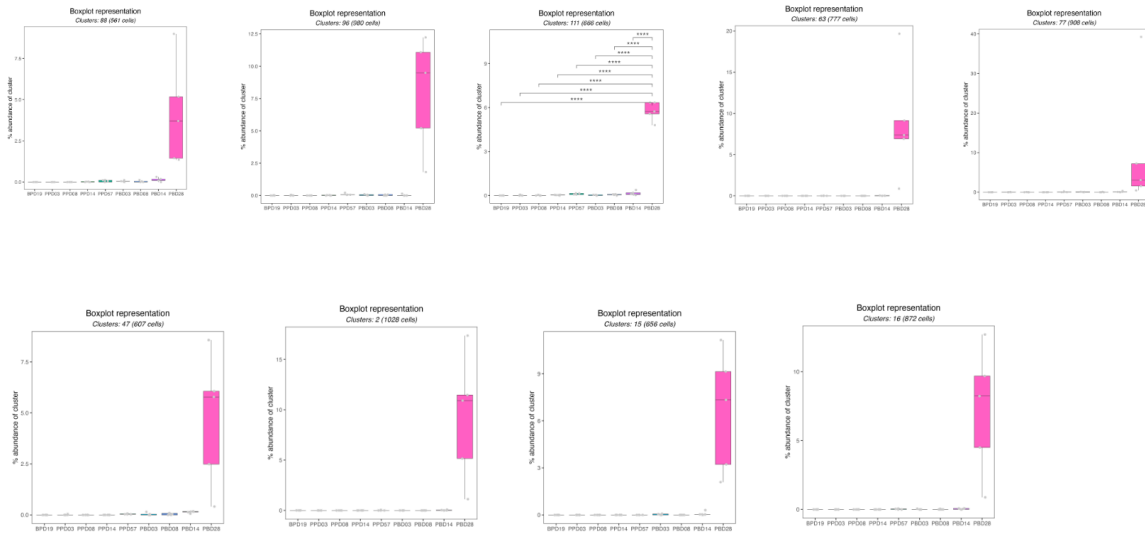


Supplementary Figure 1. Marker expression overlays on t-SNE maps reveal phenotypic and functional structure of T cells across vaccination timepoints. t-SNE projections faceted by timepoint/condition, with cells colored by arcsinh-transformed expression of representative lineage/functional markers (CD8, CD3), activation markers (CD69, HLA-DR), cytokine (IL-2), co-stimulatory receptors (CD28, CD278/ICOS), and cytotoxic effector molecule (Granzyme B). Color scales indicate relative expression intensity within each panel, enabling visual comparison of marker-enriched regions and their apparent variation across timepoints.



Supplementary Figure 2. k-means clustering and marker expression of distinct T-cell regions. (A) UMAP visualization of single T cells with k-means clustering ($k = 100$) performed in manifold space. (B) Heatmap showing median expression of lineage, activation, co-stimulatory, proliferation, cytokine, and cytotoxic markers across all clusters. Hierarchical clustering of clusters based on marker expression reveals phenotypically coherent groups that correspond to the UMAP-defined regions, supporting the biological relevance of spatially segregated clusters.

REGION 3



Supplementary Figure 3. Longitudinal stabilization kinetics of Region III metacluster populations. Boxplots show longitudinal abundance dynamics of Clusters 2, 16, 47, 15, 63, 77, 88, 96, and 111 across all timepoints. These clusters exhibit progressive enrichment and stabilization during post-prime and post-boost phases, with sustained abundance at late memory timepoints (PBD28). Cluster 96 demonstrates the strongest late-phase accumulation, while Cluster 111 shows biphasic kinetics with contraction followed by secondary post-boost enrichment. These coordinated dynamics are consistent with differentiation and stabilization of survival-associated effector-memory and memory populations within the Region III metacluster.